



Morphological characterization of solute flow in a brown earth grassland soil with crane fly larvae burrows (leatherjackets)

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ABSTRACT

Crane fly larvae, often known as leatherjackets, are a well known horticultural pest, damaging garden lawns, plants and sporting venues. They burrow into the soil and feed on the roots of plants. However, little is known about the hydrological role of leatherjackets. Such information is needed by modellers to improve water and solute models. This paper investigates flow through leatherjacket burrows in grassland fields. It compares infiltration rates, patterns of dye tracer and pore size distributions in a grassland system with and without leatherjackets. It also compares these distributions to those in an adjacent cropped tillage system which was largely devoid of leatherjackets. Dye tracer distributions show the leatherjacket burrows to act as active preferential flowpaths for water during percolation. A significant relationship between leatherjacket density and infiltration rate was found which took the form of an exponential curve. Leatherjackets were predominately found in the upper 30 cm of the grassland where there was a significantly greater volume of pores that were greater than 2 mm than in the arable soil and where visible macropores, including those open to the surface were significantly more common, covered a significantly larger cross-sectional area and were more circular in cross-sectional area. Leatherjacket burrows significantly reduced the spread of water into the surrounding soil and thereby acted as effective conduits for water to bypass the soil matrix in the upper 30 cm of the grassland. Mean dye coverage was greater at between 30 and 36 cm in the grassland soil profile, where leatherjacket burrows ended, than any other parts of the soil profile deeper than 8 cm. Such dye distributions were rare in the arable soils where very few leatherjackets were present. In some regions fallow grassland is increasing in area in response to environmental policy. Therefore, since infiltration significantly increased with leatherjacket density, consideration of their role in macropore flow will be necessary to understand the full range of hydrological and water quality implications of conversion from arable to fallow grassland management.

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1. Introduction

Despite their small spatial role, macropores can have a significant impact on infiltration and runoff (Weiler and Naef, 2003; Jarvis, 2007; Dekker and Ritsema, 1996; Leonard et al., 2001) and play a large role in the rapid transmission of water, known as preferential or bypass flow, in both the topsoil and subsoil (Franklin et al., 2007). It is well known that macropores can also exert a significant influence on solute movement (Beven and Germann 1982; McCoy et al., 1994; Heppell et al., 1999; Allaire et al., 2002). They can promote rapid, preferential transport of water and chemicals through the soil, not only because of their size but also because they are connected and continuous over sufficient distances to bypass agriculturally and environmentally important soil layers (Beven and Germann, 1982; Heppell et al., 1999). Alaoui and Helbling (2006), for example, showed that a macropore volume representing only 0.23–2% of total soil volume accounted for 74–100%

of total water flow while Lin et al. (1996) reported that 10% of pores >0.06 mm contributed about 89% of total water flow. However, for a given soil, preferential flow may vary considerably depending on soil water saturation (Lin et al., 1997; Poulsen et al., 2002). Typically, in soils under field conditions, preferential flow will be dependent upon rainfall intensity and duration allowing conditions close to saturation to be reached. Under unsaturated conditions, macropores can sometimes act as a hydraulic barrier rather than a preferential conduit for water (McCoy et al., 1994). However, full saturation is not required for macropore flow to occur but water pressures must reach close to saturation (> –10 cm) to generate non-equilibrium flow and transport in soil macropores (Jarvis, 2007). This does not mean that the soil profile needs to be entirely saturated since macropores may provide conditions at pressures close to atmospheric with rapid bypass flow contained within a much drier surrounding soil matrix (Lin et al., 1997, 1996; Jarvis, 2007).

Macropores can be formed by soil fauna (Weiler and Naef, 2003; Bouche and Al Addan, 1997), plant roots (Holden, 2005) and cracking (Haria et al., 1994). Crane fly larvae (*Tipulidae*), often known as

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leatherjackets, are a well known horticultural pest, damaging garden lawns and plants (Pesho et al., 1981). They burrow into the soil and feed on the roots of plants. In some regions they have been blamed for problems in horse racing grounds and golf courses (Simard et al., 2006) because they damage grass roots which can result in surface vegetation becoming unstable when trod on. However, virtually nothing is known about the hydrological role of leatherjackets. Most work on faunal macropore generation has focussed on the role of earthworms (Zachmann et al., 1987) or small mammals (Zaitlin et al., 2007) which have been shown to contribute significantly to the movement of water into the subsoil (Friend and Chan, 1995). If leatherjacket infestations do result in more soil burrows then they may lead to enhanced bypass flow, at least in the topsoil that they inhabit.

Arable cropland soils are less likely to have leatherjacket infestations because they are often ploughed and chemically treated. Agricultural tillage can reduce surface macropore connectivity by breaking up the soil structure (Dao, 1993). However, across Europe, and many other areas, agricultural buffer zones have recently been promoted to protect watercourses from agricultural pollution (Leeds-Harrison et al., 1999; Puckett et al., 2004). Typically these consist of strips of grassland which are left free of crops and tillage that are located at the bottom of slopes or close to watercourses. At the same time, reductions in agricultural subsidies and drives to improve river water quality such as the EU Water Framework Directive may also result in more land being taken out of agricultural production and being converted to grassland (Holden et al., 2007). There is, however, a feedback process that needs to be considered. Undisturbed soils often have more flowing macropores than arable soils because many macropore-forming soil fauna are adapted better to the undisturbed conditions and because ploughing can often remove macropore connectivity (Lachnicht et al., 1997; Shipitalo and Butt, 1999; Wuest, 2001; Vanden Bygaart et al., 2001). Enhanced bypass or subsurface flow (piping) is more likely to occur through some grass-covered slopes set-aside from agricultural production. Additionally it is known that egg and early survival for leatherjackets are favoured by moist soil conditions associated with prolonged damp weather (Lauenstein, 1986) or irrigation of pastures (Jackson and Campbell, 1975; Pesho et al., 1981) during late summer and early autumn. Thus wetter grasslands near watercourses that have been set-aside may be good sites for leatherjackets and possible enhancement of bypass flow.

This paper aims to investigate flow through leatherjacket burrows in grassland fields. It compares patterns of dye tracer and pore size distributions in a grassland system with and without leatherjackets. It also compares these distributions to those in an adjacent cropped tillage system with and without crane fly. We hypothesise that i) leatherjacket burrows act as agents for bypass flow in the topsoil and ii) bypass flow will be greater where leatherjackets are more densely populated.

2. Methods

Data were collected from Headley Hall Farm, Tadcaster, in northern England at 53°52' N, 1°20' W. The soils are of the Barkston Series, a brown earth, sandy loam soil with a high base status. No particular standard soil characteristics such as pH, soil texture or organic matter content, are known to be preferable to leatherjackets (McCracken et al., 1995) and so this soil type was suitable for the tests. The parent material is boulder clay, with limestone residue and sandy millstone grit. The texture is sandy loam in the upper soil layers becoming finer below 30 cm depth (sandy clay loam) and the soils have a mean depth of 61 cm. The surface structure is of a moderately developed crumb or fine subangular blocky nature. The soil is slightly acidic and is well suited to arable crops and grasses. On the farm we investigated sites that had been converted from arable to fallow grass for over 5 years ('grassland'). These were compared to sites which had been used for arable crops for at least the last 5 years prior to the experiment ('arable'). The latter were planted with winter wheat. Data were

collected in May and June 2007. Adult crane flies in Europe emerge from pupae in August and September. Females then lay eggs in the soil. These hatch 10–14 days later and the larvae feed through the year until they pupate in August and September. Winter wheat is typically planted in October and November following tillage and harvested in July.

Eight vertical soil cores of 12 cm diameter and 60 cm length were randomly taken from each land use for determination of bulk density and organic matter profiles at 10 cm depth intervals using standard methods (i.e. loss on ignition; Ben-Dor and Banin, 1989). Pore size distributions were estimated through use of a suction membrane device with hanging water column for eight soil samples per land use also at 10 cm depth increments. Within each land use, earthworms and leatherjackets were counted for three 50 × 50 cm plots for both 0–30 cm and 30–60 cm soil depths.

Brilliant Blue FCF was applied to the surface at 50 points using an infiltrometer with a disk of 14 cm diameter, at a tension of 0 cm, and with dye used at a rate of 5 g l⁻¹ which is within the range for good visibility (Flury and Fluhler, 1994). Each experiment was run for 30 min and final infiltration rates were measured as the value over the last 5 min of the experiment. There were 25 dye tracing infiltration experiments in each land use category. Brilliant Blue FCF has been previously recommended for soil studies by several authors (e.g. Weiler and Naef, 2003; Kim et al., 2004) and has low toxicity and high visibility (German-Heins and Flury, 2000). Its large molecules (Wang et al., 2002) have a high molecular weight which reduces lateral diffusion (Smettem and Trudgill, 1983). The adsorption of Brilliant Blue FCF in soil is dependent on soil type, with enhanced adsorption in clays and some preferential flow paths may not be completely visible since the major part of adsorption proceeds very rapidly (Ketelsen and Meyer-Windel, 1999). However, for visualizing flow patterns it is generally accepted that the dye is well suited.

Before the experiments were conducted, most of the surface vegetation was removed so that surface conditions for all plots were similar to ensure that microtopography and surface vegetation did not affect resulting dye transport (Yasuda et al., 2001). This was done by carefully using a sharp knife to cut the vegetation back close to the soil surface so that disturbance did not affect macropores below. A layer of fine sand was also placed between the infiltrometer disk and the soil surface to aid contact. Because absorption of the dye is time dependent (German-Heins and Flury, 2000) it was important to keep the time of each dye observation constant.

Two hours after the infiltration experiment had taken place the dyed area was carefully excavated and photographs of the vertical soil profile were taken using a digital camera with 5 mega pixels. Firstly an area outside of the dyed disk was dug. The pit was then extended backwards incrementally into the dyed area, revealing the vertical profile of the dye. The pits were dug to the base of the soil profile, usually around 60 cm depth. Five different vertical profiles were photographed at each infiltration site. The photographs were taken in shade so that shadow effects at the bottom of the pit would have less of an effect on the images. In addition to photographs, the orientation of visible macropores was noted along with their diameter and length using callipers where possible. In each of the pits the number of excavated leatherjackets and earthworms were counted. Additionally, the number of surface visible macropores was surveyed in 15 random 1 m² plots per land use category.

Photographic images were processed to determine the percentage of dye cover over the whole image (Forrer et al., 2000; Ohlstrom et al., 2002). Adobe Photoshop software was used on the relevant section of each image to perform a level adjustment and shadow reduction to increase the contrast of the image. The red channel was then selected as this showed the greatest contrast between the dye and surrounding soil. The red channel image was then converted to a high contrast black and white image and the threshold function (set to 128) was used to convert the dyed areas to black and soil areas to white. Any

remaining shadows that still showed up on the image as dark areas were manually selected and changed to white. From this final image the percentage of the area that was dark was determined from the image histogram. In other words the percentage of the vertical soil face that was dyed could be determined electronically from the digital images.

3. Results

The median dry bulk density of the arable land was 1.45 g cm^{-3} which was significantly greater than that for the adjacent grassland at 1.10 g cm^{-3} . However, the grassland had a much larger range of bulk density (0.93 g cm^{-3} compared with 0.42 g cm^{-3} for arable) while both land use categories had a maximum bulk density of 1.70 g cm^{-3} . Bulk density did not significantly change with depth for either treatment, although bulk density increased slightly with depth. The organic matter content of the arable sites (median 1.4%) was significantly lower ($p < 0.001$) than that in the grassland sites (2.6%) and had a smaller range of values (0.3%) compared to 1.1% for grassland).

The mean density of earthworms measured in the $50 \text{ cm} \times 50 \text{ cm}$ plots was 35 m^{-2} in the arable topsoil (upper 30 cm) and 33 m^{-2} in the subsoil while the mean density of earthworms in the grassland topsoil was 48 m^{-2} and 51 m^{-2} in the subsoil. The main earthworm species found were *Aporrectodea caliginosa*, *Aporrectodea longa* and *Octolasion cyaneum* with occasionally some *Lumbricus terrestris*. The mean density of leatherjackets measured in the $50 \text{ cm} \times 50 \text{ cm}$ plots was 0 m^{-2} in the arable soil, 37 m^{-2} in the grassland topsoil and 4 m^{-2} in the grassland subsoil. Leatherjackets were only found in one of the 25 photograph excavation pits in the arable sites but were in 18 of the 25 pits in the grassland where the maximum recorded surface areal density was 82 m^{-2} measured across the whole soil depth.

Table 1 provides the mean pore size distributions for the grassland and arable soil. A *t*-test ($p < 0.001$) showed that the proportion of the soil pore volume that was $> 2 \text{ mm}$ in diameter was significantly greater in the upper 30 cm (topsoil) of the grassland when compared to the subsoil. This proportion was also significantly greater than in the arable soils. This significant difference also resulted in a concomitantly reduced proportion for the smallest pores examined ($< 0.125 \text{ mm}$) when the grassland topsoil was compared to the subsoil or the arable soil. The only other significant difference found was for pores in the range of $0.25\text{--}0.5 \text{ mm}$ where the pore volume was greater in the arable topsoil than the arable subsoil.

Table 2 outlines some of the visible (large) macropore characteristics measured in the field. More macropores were visible at the surface and macropores were larger in the grassland where the leatherjackets were present than compared to the arable sites. Macropores were often vertically orientated with a mean orientation of 11° from the vertical. The grassland macropores were characterised by smooth rounded faunal burrows created by leatherjackets (Fig. 1A). This resulted in a macropore elongation ratio somewhat less than those created by desiccation at the arable plots (Fig. 1B). In other words, the macropores

Table 2

Characteristics of measured macropores.

Macropore characteristics	Grassland	Arable
Number of surface visible macropore openings, m^{-2}	49	11
% area of surface visible macropore openings	0.31	0.01
Mean diameter, mm	9	3
Mean length, mm	152	81
Mean angle, degrees*	11	10
Elongation ratio ⁺	1.1	1.9

*Between 0° and 90° , where 0° is vertical and 90° is horizontal.

⁺Maximum diameter of macropore divided by minimum diameter.

were more rounded and cylindrical in the grassland. The arable site surface macropores were characterised more by narrow vertical desiccation cracks such as that shown in Fig. 1B.

Large differences were observed between the two ground covers in terms of total vertical area of dye coverage ($p < 0.001$). As five vertical profiles were imaged for each experimental plot the median value for each plot was taken and then means derived from these 25 medians for each land cover type. The arable plots had a mean dye coverage in their pit profiles of 74.3% (SE = 1.9%) while the grassland plots had a mean dye coverage of 39.3% (SE = 4.1%). The grassland soil profiles were subject to a much less homogeneous dye coverage suggesting that a greater proportion of flow was taking place down larger preferential flowpaths than in the arable plots. There were frequent fingers of dye which appeared to follow the direction of leatherjacket burrows and which after about 30 cm depth would end in a bulb of dye extending from the bottom of the burrow as dye spread out from this location. Fig. 2 shows the mean dye profile with depth and two typical example profiles from each land use. The percentage dye cover in the very upper parts of the soil profile are approximately the same between the two land uses as would be expected, but the dye cover in the grassland decreases much more rapidly with depth over the upper 10 cm as the dye becomes more concentrated in preferential flowpaths. Such a sudden decline with depth in the upper soil profile is not seen in the arable soils. There is then a second peak of mean dye cover at between 30 and 36 cm depth in the grassland soil before dye coverage reduces comparatively slowly. In contrast such a secondary peak is not seen in the arable dye curves with depth. The second peak in dye cover in the grassland soil corresponds to the depth where the leatherjacket burrows appeared to cease.

While there were some desiccation cracks in the arable soils that typically penetrated from the surface to around 5–10 cm depth, the dye imaging showed that these cracks seemed to have a much smaller influence on water movement than the leatherjacket burrows in the grassland. Frequently dye was uniformly transmitted through the soil profile even where the cracks were present in both the arable and grassland soils.

As other processes under the two land use types (rooting, tillage, earthworms etc) may partly contribute to differences in dye penetration and macropore formation, the effects of the leatherjackets need to be more isolated for analysis. In order to do this it is possible to: (i) compare the seven grassland dye coverage plots where no leatherjackets were found with the 24 arable plots where no leatherjackets were found; (ii) compare infiltration rates for the sites detailed in (i) and (iii) examine the relationship between infiltration rate and leatherjackets density across the whole dataset. For plots without leatherjackets the mean vertical dye coverage for the grassland was 69%, compared to 74% for the arable land. A Mann–Whitney *U*-test showed that there was no significant difference between grassland and arable dye penetration for sites without leatherjackets ($p = 0.15$). Similarly, final infiltration rates were not significantly different for arable and grassland plots without crane fly ($p = 0.98$). Thus the adjacent grassland and arable sites were similar when leatherjackets were not present and earthworm or other faunal activity could not be attributed to the different dye patterns found.

Table 1

Pore volume distributions for the grassland with leatherjackets and the adjacent arable land without leatherjackets. Upper 30 cm of soil is classified in the table as topsoil and the lower 30 cm is subsoil.

Diameter, mm	Grassland		Arable	
	Topsoil	Subsoil	Topsoil	Subsoil
< 0.125	95.10* ⁺	96.04*	95.99 ⁺	96.36
0.125–0.25	2.20	2.02	1.98	2.11
0.25–0.5	1.64	1.75 ⁺	1.84*	1.37* ⁺
0.5–1	0.05	0.09	0.10	0.11
1–2	0.09	0.05	0.07	0.04
> 2	0.92* ⁺	0.03*	0.02 ⁺	0.01

*Significant difference ($p < 0.05$) between topsoil and subsoil, ⁺significant difference between grassland and arable.

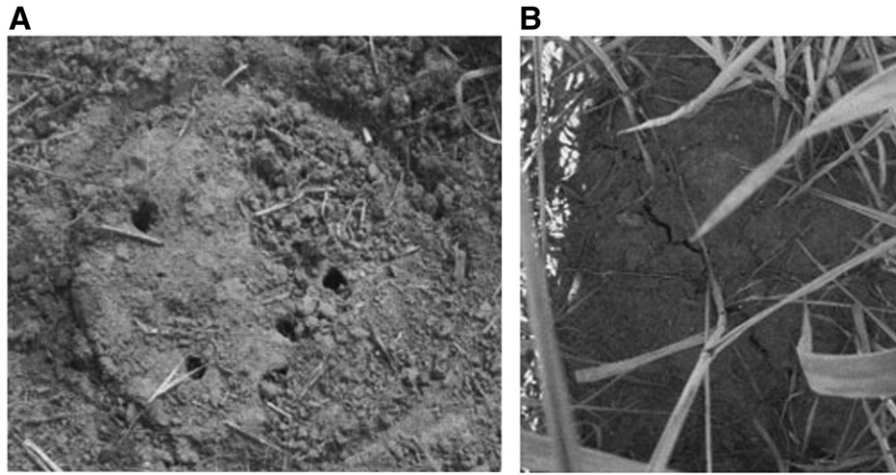


Fig. 1. Typical surface visible macropores in A) the grassland where leatherjackets were common B) the arable fields.

Final infiltration rates ranged from 4.7 mm h^{-1} to 1211.0 mm h^{-1} . For plots with crane fly the mean final infiltration rate was 406.1 mm h^{-1} while the mean rate was almost an order of magnitude smaller for plots without crane fly larvae at 49.5 mm h^{-1} . Fig. 3 shows the significant relationship ($p < 0.001$) between leatherjacket density and final infiltration rate which takes the form of an exponential curve. Included in Fig. 3 are values for plots where no leatherjackets were found. Importantly, there was no significant relationship ($p = 0.22$) between earthworm density and leatherjacket density in the dye plots or the $50 \text{ cm} \times 50 \text{ cm}$ plots either when taken as a whole or when subdivided into topsoil or subsoil layers.

4. Discussion and conclusions

Our data suggest that leatherjackets influence the morphology of bypass flow and the infiltration rate. Comparing infiltration and dye coverage data between plots on grassland without leatherjackets with arable plots without leatherjackets showed no significant differences. This suggests that it is the presence of leatherjackets that can partly explain the major differences found more generally between dye penetration and infiltration in adjacent arable and grassland sites since the grassland was dominated by leatherjacket burrows and they were rare on the arable land. Additionally the significant differences in pore size distributions and visible macropores between arable and grassland were probably, at least in part, related to leatherjacket burrowing.

Data on soil pore distributions showed that the grassland had a significantly greater volume of pores that were greater than 2 mm, particularly in the topsoil, than the arable soil. At the same time, visible

macropores, including those open to the surface were significantly more common, covered a significantly larger cross-sectional area and were more circular in cross-sectional area in the grassland compared to the adjacent arable field. Leatherjackets were rarely found in the arable fields but were found in large numbers in the adjacent grassland. Leatherjackets occurred predominantly in the upper 30 cm of the soil profile in the grassland while earthworms occurred in equal amount in the top and bottom 30 cm of the grassland.

There were clear differences in dye tracer distributions in the grassland and adjacent arable soil which support the idea that the leatherjacket burrows act as dominant routes of preferential flow. The pattern of dye tracer distribution in the grassland soils was similar to that found by Alaoui and Goetz (2008). Their data suggested this pattern occurred where large macropores ended and the macropore network became smaller and denser allowing more dye to penetrate into the soil matrix. This mechanism would explain the patterns of dye observed in the grassland site. Dye is able to bypass much of the matrix in the upper 25 cm of the soil by flowing through large macropores which then end allowing dye to be transmitted into the matrix at this point and hence to spread out through the soil profile increasing the spatial coverage of dye. The locations of the leatherjackets and their burrows would suggest that they played an important role in such dye tracer distributions.

Greater infiltration and macroporosity on grassland soils when compared to arable soils has been previously attributed to greater organic matter contents and a potentially greater earthworm content (Dao, 1993; Tebrugge and During, 1999). Our data shows that earthworm contents are indeed 30% greater in the grassland soil than the

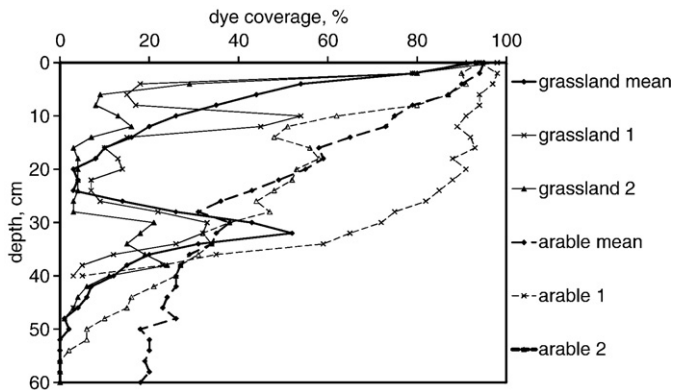


Fig. 2. Mean dye profile (thick lines) and two typical dye profiles with depth for each land cover.

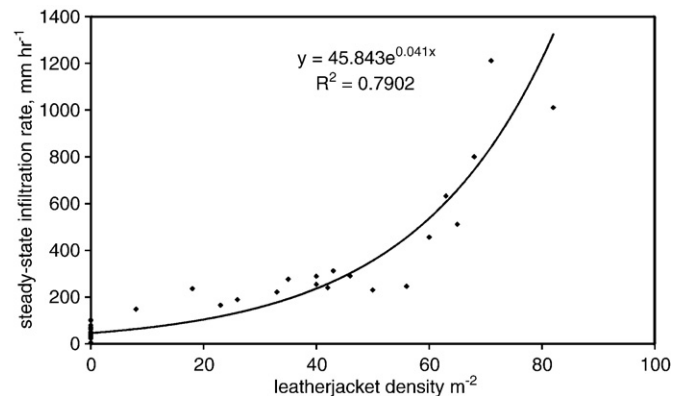


Fig. 3. Leatherjacket density against steady-state infiltration rates for each sampling plot.

adjacent arable soil, but there was little difference between topsoil and subsoil worm density in the grassland (48 m^{-2} c.f. 51 m^{-2}). Crane fly larvae, however, were generally restricted to the topsoil (37 m^{-2} in the grassland topsoil and 4 m^{-2} in the grassland subsoil). So while earthworm activity could explain a small part of the differences in dye distribution between the arable and grassland sites they cannot account for the main observed patterns which were more clearly related to differences in faunal activity in the topsoil. Furthermore, the fact that there was no significant relationship between earthworm density and leatherjacket density in the plots while there was, for example, a significant relationship between leatherjacket density and infiltration, adds weight to our evidence that the presence of leatherjackets has a major impact on macropore flow and infiltration.

Roots of winter wheat can extend quite deep and they were found throughout the soil profile to its base at around 60–70 cm. They may therefore aid deep preferential flow development, particularly at maturity which was at the time of our sampling. However, bypass flow as indicated by dye coverage and by infiltration data was found to be greater in the grassland sites with leatherjackets and there were no differences in dye coverage and infiltration between plots without leatherjackets when compared between arable and grassland sites. There were differences in the nature of the bypass flow found in the arable plots which appeared to be related to desiccation cracks. Desiccation cracks were common in the arable plots sampled, but were uncommon on the grassland plots. The greater organic matter content in the grassland soils will have reduced the swelling and shrinking (Hillel, 2004) and therefore reduced the likelihood of desiccation crack formation in the grassland soils. Many studies have examined preferential flow down desiccation cracks but most of these studies have been focussed on clay soils (Brown et al., 1999; Heppell et al., 2002) or peat soils (Holden, 2005; Holden et al., 2001) which are more susceptible to shrinkage and swelling and hence more prone to cracking. Field evidence suggested that in the arable sandy loam soils tested, the smaller cracks may have been destroyed by swelling during wetting and some of the larger ones were filled with debris which impeded flow. Frequently dye was uniformly transmitted through the soil profile even where the cracks were present. This may be related to a wetting process whereby water fills the cracks and then infiltrates laterally into the surrounding soil matrix (Hillel, 2004). In the grassland, this pattern was only seen at the base of leatherjacket burrows. The dominant process of flow in the arable soils seemed to be the advancement of the saturated wetting front, which created uniform dye coverage throughout the upper soil profile which decreased rapidly with depth. However, some flow via cracks was apparent.

Bypass flow is greater where leatherjacket density is greater. We found a significant exponential relationship between leatherjacket density and final infiltration rate. This is an important finding for understanding solute and water flow in areas where crane fly are present. However, it should also be noted that we do not expect the exponential relationship to continue much beyond the range of data we have measured. This is because at greater densities of leatherjackets the macropores created by burrowing will begin to interact so that the rate of increase in infiltration rates will not be as great. Furthermore, an alternative interpretation of Fig. 3 is one based on two thresholds. Towards the origin of the plot, the presence of leatherjackets is the first threshold since the steady-state infiltration rate for low densities is always greater than that at zero density of leatherjackets. For low densities of leatherjackets, but where there are leatherjackets present (i.e. non zero densities), the steady-state infiltration rate can then be seen to be fairly constant up to a leatherjacket density of 57 m^{-2} . Above this second threshold density the infiltration rate rapidly increases. This increase in steady-state infiltration rate with leatherjacket densities above the threshold of 57 m^{-2} could be interpreted as a linear increase. Further work is required to establish whether such thresholds exist or alternatively whether exponential relationships exist between leatherjacket den-

sity and infiltration rate or other hydrological properties for different environmental conditions or soil types. The empirical curve we have derived could be of significant benefit to water and solute modellers, either as an input for simplified infiltration models or as an output of further processed-based models aiming at better understanding the role of leatherjacket macropores.

Clearly, more work is needed for a wide range of soil types and environments but our paper provides a necessary first step. It should also be noted that our experiments may not be representative of natural rainfall conditions and the high infiltration rates measured on site are unlikely to be exceeded by rainfall intensities (at least over long durations) in the humid temperate zone except on rare occasions. Nevertheless, the experiments do indicate that leatherjackets influence flow pathways within soils and have an impact on the infiltration properties and the active macropore structure. Given the patterns of flow indicated by the dye tracing analysis it is also likely that leatherjacket burrows will influence lateral throughflow processes in the upper soil layer.

While many arable farming methods reduce leatherjacket presence to a minimum, larvae burrows may increase dramatically if arable land is set-aside from production and left fallow. This may have significant impacts on infiltration, water flow and solute production processes. Modelling work will be needed to aid prediction and understanding of impacts and the form of relationships we have found between leatherjacket density and infiltration will be a useful addition.

While it is often assumed that arable farming increases the rate of removal of water from the land to watercourses (Holden et al., 2007), this may therefore not necessarily be the case. Under current farming conditions in Europe, for example, more land is being set-aside from agricultural production and left as grassland. This has been a reaction to falling agricultural subsidies and the need to protect watercourses from agricultural pollution. Hence many fallow fields are left adjacent to watercourses to provide buffer zones. However, undisturbed soils are more likely to have higher populations of soil macro-organisms and these may produce more continuous macropore structures that transmit more bypass flow. This may have undesired results for some pollutants that the grassland buffer strips are meant to be protecting watercourses against. This could be a critical area for further research in order to maximise the potential improvements in water quality that buffers could introduce. Of course, the effects may only be significant if the macropores are sufficiently connected to some outlet for the water from the soil system such as field underdrainage or other macropore networks. Nevertheless, further experiments will be required to isolate the role of leatherjackets and investigate their distributions and hydrological role. Further work is also required in a wider range of soil types and surface cover types where leatherjackets are common.

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